

THE GRAIL COLLECTOR

Dataset Audit Addendum

An independent engineering and CFD audit of GRAIL_CFD_dataset_rev2.csv, and the eight reporting corrections it requires.

Item	Value
Dataset audited	GRAIL_CFD_dataset_rev2.csv, 300 rows x 54 columns
Method	Independent recomputation of every derived column from its inputs
Data modified	NONE — no row deleted, corrected, smoothed or replaced
Critical engineering errors found	0
Conservation-law violations found	0
Corrections required	8, all in reporting and nomenclature
Results invalidated	0

Companion to the CFD Simulation Report

Contents

(In Word: right-click the field and choose Update Field.)

1. Purpose and method

This addendum records an audit of the reported dataset carried out to a single rule: verify everything, change nothing. Every derived column was recomputed independently from its own inputs and compared against the stored value. No row was deleted, corrected, smoothed, replaced or reweighted, and no unusual result was altered because it was unusual.

The audit found no critical engineering error and no violation of any conservation law. It found eight items requiring correction, and every one of them is a correction to how a result is DESCRIBED, not to the result itself. This document states each one, gives the dataset evidence, and says exactly what changes in the reporting.

1.1 What was verified exactly

The following relations were recomputed from first principles across all 300 rows. They are not merely plausible — they close to machine precision, which means the dataset is algebraically self-consistent.

Table 1 — Independent verification of every derived relation.

Relation verified	Maximum error over 300 rows
$\dot{m}_{\text{total}} / \dot{m}_{\text{channel}} = 12$ exactly	0
$q'' = G_T \times 0.833 \times 0.82 \times 0.95$	$1.1\text{e-}13$
$Q_{\text{solar}} = q'' \times 0.528 \text{ m}^2$ (1.1×0.48)	$5.7\text{e-}14$
$Q_u = \dot{m}_{\text{dot}} \times c_p \times dT$, $c_p = 4180$	$< 1\text{e-}6 \%$
$dT = T_{\text{out}} - T_{\text{in}}$	$1.1\text{e-}13$
$Q_{\text{solar}} = Q_u + Q_{\text{rad}} + Q_{\text{conv}} + Q_{\text{rear}}$	$1.43\text{e-}04 \text{ W} = 9.83\text{e-}05 \%$
$Re = 4 \dot{m}_{\text{dot}} / (\pi D_h \mu)$	$8.0\text{e-}13$
$\mu(T)$, Vogel relation	$2.3\text{e-}13$
$W_{\text{pump}} = dp \times \dot{m}_{\text{dot}} / \rho$	$9.5\text{e-}13$
$D_{h,\text{out}} = D_{h,\text{in}} \times G$; $A_{\text{out}} = A_{\text{in}} \times G^2$	$1.4\text{e-}14$
$T_{\text{sky}} = 0.0552 T_{\text{amb}}^{1.5}$; $h_{\text{wind}} = 5.7 + 3.8 v$	$1.7\text{e-}13$
$\eta = Q_u / (G_T \times A)$	$2.2\text{e-}16$
The energy balance is real, not self-referential	
The stored energy_error_pct column was not taken on trust. The balance was recomputed from the four component columns Q_u , Q_{rad} , Q_{conv} and Q_{rear} against Q_{solar} , and the recomputation reproduces the stored value to the last digit.	
Maximum residual across 300 cases: $1.43\text{e-}04 \text{ W}$, which is $9.83\text{e-}05 \%$ of the absorbed power.	
No conservation law is violated anywhere in the dataset.	

Eight physical ordering tests were also applied, with zero violations in 300 rows: $T_{\text{max}} \geq T_{\text{mean}} \geq T_{\text{min}}$; $P90 \leq P95 \leq P99 \leq T_{\text{max}}$; $\text{spread} = \text{max} - \text{min}$; $\text{std} \leq \text{spread}$; $\text{mean}(T^4) \geq \text{mean}(T)^4$ (Jensen's inequality); $T_{\text{R4}} = R4^{0.25}$; $T_{\text{out}} > T_{\text{in}}$; and $dp > 0$.

2. Item 1 — nu_{cfd} is a fixed closure, not a per-case CFD result

Classification: MAJOR METHODOLOGICAL. This is the most important item in the audit.

2.1 The evidence

Table 2 — nu_{cfd} against the range of conditions it is applied over.

Quantity	Range across the 300 cases
nu_{cfd}	2.9238 on every row — exactly one distinct value
Re, inlet	29.1 to 200.0
Re, outlet	36.3 to 540.4
Prandtl number	2.41 to 6.84 — a factor of 2.84
Mean fluid temperature	293.3 to 353.0 K
Grading ratio G	0.351 to 0.998

For laminar internal flow, $Nu = f(Re, Pr, Gz, \text{section shape})$. A quantity genuinely extracted from each case's own CFD solution could not be constant to five significant figures while the Prandtl number alone changes by a factor of 2.84.

2.2 What it actually is

The value 2.9238 is the Richardson extrapolation of the three-level 3-D grid study (3.7397 at 34,304 cells, 3.4282 at 105,600, 3.2236 at 356,400; observed order $p = 1.283$, GCI 11.63 %). It is a single developed-region constant applied as a closure to every case. That is a legitimate modelling choice. Presenting it under a column name that implies per-case CFD extraction is not.

2.3 What changes

1. The column is to be described everywhere as $nu_{closure}$ — a fixed Nusselt closure taken from the resolved 3-D CFD — and never as a per-case CFD output. The CSV column name is left unchanged so that existing scripts and figures still run; the data dictionary carries the correction.
2. Every table and figure derived from the campaign must state that a constant Nu is carried.
3. The cost of the assumption is already quantified and must be quoted alongside the result: carrying the measured $Nu(x_i)$ profile instead of a constant moves the plate-spread difference from -21.09% to -6.85% at matched developed-region mean, a shift of $+14.24$ percentage points, which is larger than every numerical and input uncertainty in the assessment combined.

What supports the assumption, from the dataset itself

The thermal entry length $x_{fd,t} = 0.05 Re Pr D_h$ was computed for all 300 cases. It ranges 0.037 to 0.163 m against a channel length of 1.100 m.

In EVERY case the flow is thermally developed over at least 85 % of the channel, and in no case does the entry region exceed 15 % of the length.

A developed-region constant is therefore a defensible first approximation on this geometry. It is still an approximation, and its cost is the $+14.24$ pp above.

Verdict on the CFD results: VALID, MODEL-FORM LIMITED. No result is invalidated. The uniformity claim must carry the model-form qualifier; the efficiency claim moves by only -0.57 pp under the same test and is unaffected.

3. Item 2 — U_L is ill-conditioned in 20 cases, and the campaign statistic is restated

Classification: MAJOR, AFFECTS A REPORTED STATISTIC. This item changes a published number.

3.1 The evidence

The definition was confirmed against the data to 1.0e-4 relative:

$$U_L = (Q_{\text{rad}} + Q_{\text{conv}} + Q_{\text{rear}}) / (A \times (T_{\text{plate,mean}} - T_{\text{amb}}))$$

The denominator ($T_{\text{plate,mean}} - T_{\text{amb}}$) spans -8.54 to +88.53 K across the campaign. Where it approaches zero, U_L diverges.

Table 3 — U_L conditioned on the size of the driving temperature difference.

Subset	Rows	U_L range (W/m ² K)
$ T_{\text{plate,mean}} - T_{\text{amb}} < 5 \text{ K}$	20	-21.86 to +25.27
$ T_{\text{plate,mean}} - T_{\text{amb}} \geq 5 \text{ K}$	280	0.850 to 3.440

The 280-row population is tight and physically sensible. The 20-row tail is a small-denominator artifact of the definition, not a property of the collector and not a defect in the CFD. The five rows with negative U_L all have a negative denominator (-1.125 to -0.257 K) with a positive numerator (1.07 to 2.98 W).

3.2 The consequence — a reported result is restated

The CFD Simulation Report quotes a campaign loss-coefficient difference of $-0.90 \pm 22.93 \%$ and marks it NOT DISTINGUISHABLE. That $\pm 23 \%$ band is now shown to be dominated by division by a near-zero temperature difference in 20 of 300 rows, not by any real uncertainty in the loss coefficient.

Table 4 — Delta U_L , alternating minus co-current, against the conditioning threshold.

Threshold on $ T_{\text{plate,mean}} - T_{\text{amb}} $	Rows	co-current	alternating	Difference
0 K (all rows, as published)	300	2.253	2.232	-0.94 %
$\geq 2 \text{ K}$	291	2.356	2.254	-4.35 %
$\geq 5 \text{ K}$ (recommended)	280	2.307	2.262	-1.95 %
$\geq 10 \text{ K}$	260	2.303	2.260	-1.87 %
The restated result				
On the conditioned subset (280 rows, $ T_{\text{plate,mean}} - T_{\text{amb}} \geq 5 \text{ K}$):				
Delta $U_L = -1.95 \% \pm 2.52 \text{ pp}$ ($k = 2$).				
The verdict is unchanged — it still spans zero and is still NOT distinguishable. But the expanded band falls from $\pm 22.93 \text{ pp}$ to $\pm 2.52 \text{ pp}$, a factor of nine.				
The statement therefore strengthens, from 'too noisy to tell' to 'genuinely no measurable difference'. That is a better scientific claim, and it costs nothing, because the conditioning removes an artifact of the estimator rather than any data.				

The 20 excluded rows are NOT deleted from the dataset. They remain in the file, they remain valid cases, and every other quantity computed from them (efficiency, useful heat, plate statistics, energy balance) is retained and used. Only the U_L statistic, which is undefined in that regime, is reported on the conditioned subset — and the threshold is stated wherever it is used.

Verdict on the CFD results: VALID. The derived quantity is ill-posed near zero driving temperature difference. No case is invalidated.

4. Item 3 — constant thermal conductivity against temperature-dependent viscosity

Classification: MAJOR, PROPERTY-MODEL INCONSISTENCY. This item adds a new term to the uncertainty budget.

4.1 The inconsistency

Viscosity is carried as a function of temperature through the Vogel relation and varies by a factor of 2.84 across the campaign. Thermal conductivity is held fixed at 0.610 W/m·K. The two properties belong to the same fluid over the same temperature range, and treating one as variable and the other as constant is not defensible without a bound on the effect.

Table 5 — The size of the inconsistency.

Quantity	Value
Mean fluid temperature across the campaign	20.2 to 79.8 °C
k(T) for liquid water over that range	0.6021 to 0.6737 W/m·K
Value used by the model	0.610 W/m·K, fixed
Resulting error in k	–1.30 % to +10.44 % (mean +5.02 %)
Rows where the error exceeds 5 %	166 of 300
Rows where the error exceeds 8 %	46 of 300

4.2 Bounding it without re-running the campaign

The heat-transfer closure is

$$h(x) = Nu \times k / D_h(x)$$

so an error in k is mathematically identical to the same percentage error in Nu. The study already contains a MEASURED sensitivity to Nu, obtained by running the entire campaign twice with nothing changed but that one input:

Table 6 — The calibrated sensitivity, from the Rev 1 to Rev 2 recomputation.

Quantity	Value
Nu changed from	3.4282 to 2.9238 (–14.71 %)
Delta plate spread changed from	–12.010 % to –21.093 % (–9.083 pp)
Local slope	0.6175 pp per 1 % change in Nu, and therefore in k

Table 7 — The k(T) term mapped onto the reported result.

Error in k	Effect on Delta plate spread	Corrected value
–1.30 % (coldest cases)	–0.80 pp	–21.89 %
+5.02 % (campaign mean)	+3.10 pp	–17.99 %
+10.44 % (hottest cases)	+6.44 pp	–14.65 %
What this means		
Using a fixed k understates the heat-transfer coefficient in the hotter cases, and by the calibrated slope this makes the reported uniformity benefit slightly MORE favourable than it should be.		
At the campaign mean, correcting k would move Delta plate spread from –21.09 % to about –18.0 %. At the extreme it would reach –14.7 %.		

Error in k	Effect on Delta plate spread	Corrected value
The sign of the result does not change, and the benefit remains clearly negative under every value in the range. This term is larger than the plate-grid discretisation term (2.98 pp) and comparable to the input-uncertainty term (4.98 pp), but smaller than the Nu(xi) model-form term (14.24 pp). It belongs in the uncertainty budget and is added there.		

Note that c_p requires no such correction. Over the actual range of 293 to 353 K, the specific heat of liquid water deviates from 4180 J/kg·K by at most +0.29 %. The constant- c_p assumption is fully justified and is not flagged.

Verdict on the CFD results: VALID, with a newly quantified uncertainty term. No result is invalidated. Required to close the item properly: re-run the campaign with $k(T)$, which costs the same as the Rev 1 to Rev 2 recomputation.

5. Items 4 and 5 — sign convention, and efficiencies above the optical limit

Classification: NOMENCLATURE AND EXPLANATION. The physics is correct in both cases.

5.1 Item 4 — negative Q_{rad} , Q_{conv} and Q_{rear}

These are not errors. Each one occurs when the relevant surface is COLDER than what it exchanges heat with, so the flux is directed into the collector. Newton's law of cooling and the Stefan-Boltzmann difference both change sign correctly; the solver is right and the word “loss” is what fails.

Table 8 — Every negative value traced to its physical cause.

Column	Negative rows	Verified cause
$Q_{\text{conv_W}}$	11 of 300	$T_{\text{plate}} - T_{\text{amb}} = -8.54$ to -2.83 K
$Q_{\text{rear_W}}$	16 of 300	same rows, same cause
$Q_{\text{rad_W}}$	11 of 300	$T_{\text{plate}} - T_{\text{glass}} = -5.21$ to -0.81 K
$U_{\text{L_W_m2K}}$	5 of 300	negative denominator, positive numerator (Section 3)

The identity of the radiative exchange was confirmed rather than assumed: the correlation between Q_{rad} and $(T_{\text{plate}} - T_{\text{glass}})$ across all 300 rows is +0.982, which establishes that the column is the absorber-to-glazing exchange and that its sign follows that temperature difference exactly as it must.

What changes: the four columns are to be described as NET FLUXES with an explicit sign convention stated once and applied everywhere — positive out of the collector, negative into it. The word “loss” is reserved for cases where the flux is genuinely outward. No value is altered.

5.2 Item 5 — eleven cases with $Q_{\text{u}} > Q_{\text{solar}}$

The optical ceiling of this collector is $\tau_{\text{glz}} \times \tau_{\text{TIM}} \times \alpha = 0.833 \times 0.82 \times 0.95 = 0.6489$. Eleven cases report an efficiency above it, reaching 0.6778, with $Q_{\text{u}}/Q_{\text{solar}}$ up to 1.0445 and a loss sum of -4.45% of absorbed power.

They are the same eleven cases as the negative-flux rows in Section 5.1. In each one the inlet temperature is well below ambient at moderate irradiance, so the absorber runs colder than the surrounding air and the collector harvests ambient heat IN ADDITION to sunlight. The efficiency denominator counts only the solar term, so the ratio legitimately exceeds the optical limit.

Table 9 — The eleven cases, with the conditions that produce them.

Case	G_{T} (W/m ²)	T_{in} (K)	T_{amb} (K)	$T_{\text{plate,mean}}$ (K)	eta	Energy error
ALT_025	421.50	298.66	310.43	306.94	0.6549	-0.0001 %
ALT_066	431.78	290.62	309.90	301.36	0.6778	-0.0001 %
ALT_113	546.39	288.27	303.99	298.85	0.6569	-0.0001 %
ALT_128	702.23	293.73	312.01	307.96	0.6560	+0.0001 %
ALT_135	512.77	289.06	310.75	303.68	0.6695	+0.0001 %
PAR_012	771.82	291.94	311.87	304.72	0.6618	-0.0000 %
PAR_083	738.01	289.37	305.52	300.28	0.6586	-0.0000 %
PAR_103	555.38	291.77	307.82	302.26	0.6621	-0.0000 %
PAR_120	464.92	292.33	310.15	303.73	0.6627	-0.0000 %
PAR_131	830.79	293.23	308.94	306.11	0.6522	-0.0000 %
PAR_148	446.44	297.62	311.30	306.40	0.6578	-0.0000 %

In every one of the eleven the plate is colder than ambient, and in every one the energy balance closes to within one ten-thousandth of a per cent. These are real operating states that a physical collector would also exhibit, not numerical artifacts.

What changes: the definition of efficiency used here must be stated explicitly as solar-referenced, $\eta = Q_u / (G_T A)$, and a footnote must accompany any efficiency table explaining that solar-referenced efficiency can exceed the optical limit when the collector also draws heat from warmer surroundings. No case is removed or capped.

6. Item 6 — lateral_bridge_W, and why bridge width barely matters

Classification: THE ACCOUNTING IS CORRECT; THE INSENSITIVITY NEEDS EXPLANATION.

6.1 It is correctly excluded from the energy balance

This was tested directly rather than assumed. The energy balance was recomputed twice, once as reported and once with lateral_bridge_W added as a fifth term:

Table 10 — Direct test of how lateral_bridge_W enters the accounting.

Energy balance formulation	Maximum residual
$Q_{\text{solar}} - (Q_{\text{u}} + Q_{\text{rad}} + Q_{\text{conv}} + Q_{\text{rear}})$	1.43e-04 W
$Q_{\text{solar}} - (Q_{\text{u}} + Q_{\text{rad}} + Q_{\text{conv}} + Q_{\text{rear}} + \text{lateral_bridge})$	290.2 W

Adding it destroys the balance, which proves it is NOT being counted as useful energy, as an external loss, or as an independent source or sink. It is an internal redistribution diagnostic — heat moving sideways from one channel's territory to its neighbour's, inside the control volume — and it is handled correctly.

Table 11 — The two populations.

Arrangement	Minimum	Maximum	Mean
Co-current	6.46e-11 W	9.42e-11 W	7.77e-11 W
Alternating	59.46 W	290.21 W	157.14 W

The co-current values are round-off around the exact zero that y-periodicity requires: with every channel flowing the same way, the plate field is periodic in y and the temperature gradient at each bridge midline vanishes identically. The contrast must therefore always be quoted as an absolute figure and never as a ratio to zero.

6.2 Why bridge width has almost no effect

A standardised regression of lateral_bridge_W on all eight sampled inputs, over the 150 alternating cases, gives $R^2 = 0.972$ and the following coefficients:

Table 12 — Standardised regression coefficients for lateral bridge conduction.

Input	Standardised coefficient
G_T	+0.792
mdot_total	-0.605
T_in	-0.197
T_amb	+0.144
v_wind	-0.032
bridge_mm	-0.008
g_ratio	+0.007
lambda_G	-0.004

Bridge width is swept from 20 to 37 mm — an 85 % change in the conduction path — and moves lateral conduction essentially not at all. This is unusual and it deserves an explanation rather than a footnote.

The explanation, and how to verify it

The Hottel-Whillier-Bliss chain computed for this absorber gives a fin efficiency of 0.9995. With 2 mm of aluminium at 229 W/m-K spanning a 31.3 mm bridge against a loss coefficient near 2.5 W/m²K, the fin parameter $m(W-D)/2$ is very small and the bond land is already effectively isothermal across its width.

When a conduction path is already near-isothermal, the heat crossing it is set by the temperature difference imposed at

its ends, not by the width of the path. Widening the bridge adds area and length in nearly equal measure.

There is a second, geometric reason. The campaign defines $w_{ch} = \text{pitch} - \text{bridge}$, so widening the bridge simultaneously narrows the channel. The two effects act in opposite directions and partly cancel.

Required verification before this is presented: either an explicit $k A / L$ scaling argument showing the cancellation, or a controlled sweep of bridge width at FIXED channel width, which the current campaign design cannot provide.

Verdict: NOT DEMONSTRATED INVALID. The dataset value stands unchanged. What is required is the explanation above, supported by one of the two verifications.

7. Items 7 and 8 — operating pressure, and convergence evidence

7.1 Item 7 — one case above the saturation temperature

Classification: DOCUMENTATION. One case exceeds the 1 atm saturation temperature of water.

Table 13 — Cases approaching and exceeding saturation.

Condition	Rows
T_out > 353 K (80 °C)	23
T_out > 363 K (90 °C)	5
T_out > 368 K	2
T_out > 373.15 K (T_sat at 1 atm)	1 — case PAR_134, T_out = 373.21 K
T_plate,max > 373.15 K	10 (peak 400.98 K)

The solver carries single-phase liquid properties with no phase-change model. The result is valid if and only if the hydraulic loop is pressurised above the corresponding saturation pressure. At 2 bar absolute, T_sat is about 393 K and every case in the dataset, including PAR_134 and all ten high plate temperatures, sits comfortably inside the single-phase envelope.

What changes: the operating pressure of the collector loop must be stated explicitly in the methodology, together with the resulting saturation temperature and the statement that all 300 cases lie within the single-phase envelope at that pressure. If the design is an unpressurised 1 atm loop, PAR_134 must be flagged in place as outside the model's validity envelope. It is NOT to be deleted under either reading.

7.2 Item 8 — convergence evidence is a single scalar

Classification: DOCUMENTATION. The evidence is sufficient for the conclusions drawn, but it is thinner than a strong CFD publication expects.

Table 14 — Convergence information the dataset carries.

Field	Content
residual	one scalar per case, 7.59e-07 to 1.00e-06
outer_iters	73 to 849; the 900 cap is never reached
converged	True on all 300 rows
energy_error_pct	maximum magnitude 9.83e-05 %

Fifty-one rows sit at the 1e-06 tolerance boundary, meaning they converged to the tolerance rather than well past it. A single outer-loop residual does not demonstrate that continuity, momentum and energy have each converged, and for a publication the per-equation residual histories and at least one monitored engineering quantity should be shown.

What the dataset does have, which is stronger

An INDEPENDENT energy balance that closes to 9.83e-05 % on every one of the 300 cases.

A residual is a statement about the linear solver. An energy balance is a statement about the physics, computed from quantities the solver did not iterate on. It is the better evidence, and it is what justifies treating these cases as converged.

Recommended addition for publication: per-equation residual histories and a monitored quantity trace for a representative subset, alongside the balance already reported.

8. Summary and final classification

Table 15 — All eight items, classified.

#	Item	Class	Result invalidated?
1	nu_cfd is a fixed closure, not a per-case CFD value	Major methodological	No — model-form limited
2	U_L ill-conditioned in 20 of 300 cases	Major, affects a statistic	No — statistic restated
3	Constant k with temperature-dependent mu	Major, property model	No — new uncertainty term
4	Negative Q_rad, Q_conv, Q_rear	Nomenclature	No — physics correct
5	Eleven cases with eta above the optical limit	Explanation required	No — physics correct
6	Bridge width does not affect bridge conduction	Unusual, needs explanation	No — not shown invalid
7	One case above 1 atm saturation temperature	Documentation	No — state the pressure
8	Single-scalar convergence evidence	Documentation	No — balance is stronger

8.1 The separation that matters

- **ACTUAL ERRORS:** none. Not one value in the 300 x 54 table is mathematically, dimensionally or physically wrong. No conservation law is violated. No derived relation fails.
- **UNUSUAL BUT VALID PHYSICS:** items 4, 5 and 6 — the inward heat fluxes, the above-optical efficiencies, and the bridge-width insensitivity. All three remain in the dataset unchanged and require explanation, not correction.
- **REPORTING AND DOCUMENTATION:** items 1, 4, 5, 7 and 8. These change how results are described, not what they are.
- **NUMERICAL CONSEQUENCE:** items 2 and 3 are the only two that change a reported number. Item 2 restates Delta U_L as $-1.95 \pm 2.52 \%$ on the conditioned subset, a nine-fold tightening. Item 3 adds a k(T) term that would move Delta plate spread from -21.09% to about -18.0% .

8.2 The headline results after the audit

Table 16 — Reported results, with every audit correction applied.

Quantity	As reported	After audit	Verdict
Delta efficiency	$-4.94 \pm 0.81 \%$	unchanged	FIRM — a deficit
Delta plate spread (RMS)	$-21.09 \pm 11.61 \%$	about -18.0% with k(T)	REAL, model-form limited
Delta loss coefficient	$-0.90 \pm 22.93 \%$	$-1.95 \pm 2.52 \%$	NOT distinguishable, now sharply so
Delta peak-to-peak spread	$-3.64 \pm 15.82 \%$	unchanged	NOT distinguishable — do not claim

NOTE (Rev 4): the bottom line below is the Rev 2 audit verdict and is kept unchanged. Its statements on the uniformity benefit and the efficiency deficit are SUPERSEDED by Section 9.

Bottom line

The audit did not find a single engineering error in the dataset, and it did not invalidate a single case.

It found three quantities described in misleading ways, two statistics needing restatement, and one property assumption needing a bound.

None of the eight items changes the direction of any conclusion. The efficiency deficit stands, the uniformity benefit stands with a slightly smaller magnitude and an explicit model-form qualifier, and the peak-to-peak claim stays withdrawn.

No data was modified, corrected, smoothed, deleted or replaced at any point in this audit.

9. Rev 4 update — what this audit's verdict becomes

This audit verified that Rev 2 is algebraically self-consistent, and that finding stands. What changed afterwards is the physics INPUT the dataset was computed with, which an internal audit cannot detect.

Audit statement	Status after Rev 4	Reason
No engineering error in the dataset algebra	STANDS	Rev 4 closes to energy error < 1.5e-4 %
Nu = 2.9238 on every row (a valid closure)	SUPERSEDED	H2-type; 3-D conjugate Nu 4.48
Efficiency deficit stands	STANDS, LARGER	Group mean -7.3 % (p = 2e-8), every flow tercile
Uniformity benefit stands, slightly smaller	WITHDRAWN	Overall -1.1 % (n.s.); +16.8 % below 2.2 g/s, -13.4 % above 3.3 g/s
Peak-to-peak claim withdrawn	STANDS	Group median spread +18.4 %
(Rev 4.1) derived columns R4/T_R4/mean_T_pow4, U_L, Re	CORRECTED	Recomputed from definitions; thermal results unchanged
Results invalidated: 0	QUALIFIED	No row deleted; Rev 2 superseded, not invalidated

Full evidence: GRAIL CFD Correction Report and CFD Simulation Report Rev 4 edition, Part V.